| | | | |
|---|---|---|---|
| 8. | Popcorn | 2.32 | 51.38 |
| 9. | Perungayum | 0.37 | 40 |
| 10. | Mushroom | 31 | 11.73 |
| 11. | Friut juice | 0.1 | 74 |
| 12. | Choclate | 0.8 | 0.09 |
| 13. | Rava | 9 | 575 |
| 14. | Biscuit | 19.7 | 498 |
| 15. | Snacks | 33.5 | 821 |

Mr. Bean wants to establish some statistical relationship between the above mentioned food contents. Here the variable under study are $X$ and $Y$ which represents the amount of Fat content and the amount of Sodium content in each food items respectively. Thus Mr. Bean gets a pair of values $(X,Y)$ for each food item. Mr. Bean further found the average fat content in all the 15 food items is $\overline{X}$ =7.8 (gms/100g) and the average sodium content in all the items is $\overline{Y}$ =325.23(mg/100g). Further, it was identified that the minimum amount of fat contained in Tropicana fruit juice is 0.1(g/100gms) and the maximum amount of fat contained in Lays is 33.5(g/100gms). Thus the fat contained in all the 15 food items is ranging from 0.1(g/100gms) to 33.5(g/100gms). Similarly, the minimum amount of sodium content contained in Kelloggys Choco is 0.09(mg/100gm) and the maximum amount of sodium content in Bhindhu appalam is 1440(mg/100gm). So, the measure of range of fat content and sodium content in all the 15 items are 33.4gm and 1439.91(mg/100gm) respectively. Besides, Mr. Bean is interested in knowing the variation of each individual item from the mean of all observations. He attempted another measure of dispersion known as standard deviation. The measure of standard deviation indicates that there is an average deviation of 11.3 (g/100gms) in fat content and 420.14(mg/100gms) in sodium content of all the 15 food items. Further, Mr. Bean is keen on finding the association between the variables $X$ and $Y$. So, the correlation analysis has been carried out. The correlation coefficient $r(X,Y)$=0.2285 indicates that there is 22.86% positive association between sodium content and the amount of fat content. Thus from this study Mr. Bean inferred and convinced that the nutrition information on the packaging of each food item is sufficient.

## Case Study-2

We collected data relating to the gold price (per gram) in two places namely Chennai Market and Mumbai Market for ten days from 20th Feb 2018 to 1st March 2018 and the same is recorded below.

| Date | 20th Feb | 21st Feb | 22nd Feb | 23rd Feb | 24th Feb |
|---|---|---|---|---|---|
| Chennai $X$ | 2927 | 2912 | 2919 | 2912 | 2921 |
| Mumbai $Y$ | 2923 | 2910 | 2907 | 2920 | 2919 |

| Date | 25th Feb | 26th Feb | 27th Feb | 28th Feb | 1st Mar |
|---|---|---|---|---|---|
| Chennai $X$ | 2921 | 2927 | 2924 | 2908 | 2893 |
| Mumbai $Y$ | 2919 | 2925 | 2918 | 2902 | 2895 |

Do we agree that the price of gold in Chennai market has its impact on Mumbai market?

Let $X$ denotes the gold price per gm in Chennai market and $Y$ denotes the gold



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